UNIT - I (COMPUTER – AN INTRODUCTION)

1.1 What is a Computer?

A **computer** is an electronic programmable machine that accepts data as input, processes it according to a set of instructions, stores the information, and produces meaningful results as output.

In simple words:

Computer is an electronic device that accepts data, processes it, stores it and produces useful information as output.

Example

Suppose a student enters:

```
Marks in 5 subjects:
80, 75, 90, 70, 85
```

The computer can:

- Accept the marks as **input**
- Calculate total and percentage as **processing**
- Store the marks as **data**
- Display total and percentage as **output**

For example:

```
Total = 400
Percentage = 80%
```

Evolution of Computers (History)

The evolution of computers refers to the development of computing machines from simple calculating tools to the highly intelligent systems we use today.

The journey of computers began with basic mechanical devices and moved forward through various stages of technological advancement.

Early Calculating Devices

Device	Inventor	Year	Description
Abacus	Chinese Civilization	~3000 BC	First manual calculator using beads.
Napier's Bones	John Napier	1617	Used for multiplication and division.

Pascaline	Blaise Pascal	1642	First mechanical calculator.
Analytical Engine	Charles Babbage	1837	Considered the first concept of a modern computer.

Generations of Computers

The development of computer technology is divided into **five generations**, each marked by significant advancements in hardware and software.

1st Generation Computers (1940–1956)

- **Technology Used:** Vacuum Tubes
- **Programming Language:** Machine language (binary code – 0s and 1s)
- **Size:** Very large (occupied entire rooms)
- **Speed:** Very slow
- **Examples:** ENIAC, EDVAC, UNIVAC
- **Drawbacks:**
    - Produced a lot of heat
    - Required air conditioning
    - Frequently failed

2nd Generation Computers (1956–1963)

- **Technology Used:** Transistors (replaced vacuum tubes)
- **Programming Language:** Assembly language
- **Size:** Smaller and more reliable
- **Speed:** Faster than 1st generation
- **Examples:** IBM 1401, IBM 7094
- **Advantages:**
    - Less heat
    - Less power consumption
    - Better performance

3rd Generation Computers (1964–1971)

- **Technology Used:** Integrated Circuits (IC)
- **Programming Language:** High-level languages (like FORTRAN, COBOL)
- **Size:** More compact
- **Speed:** Much faster
- **Examples:** IBM System/360
- **Advantages:**
    - Lower cost
    - Better storage and processing
    - Supported multitasking

4th Generation Computers (1971–2010)

- **Technology Used:** Microprocessors (first was Intel 4004)
- **Programming Language:** All modern programming languages
- **Size:** Very small – personal computers
- **Speed:** Very high processing speed
- **Examples:** Desktop computers, laptops, tablets
- **Advantages:**
    - Affordable for personal use
    - Highly reliable
    - Use of GUI (Graphical User Interface), mouse, keyboard

5th Generation Computers (2010 & Beyond)

- **Technology Used:** Artificial Intelligence (AI), robotics, cloud computing, quantum computing
- **Programming Concepts:** Machine learning, Natural Language Processing (NLP)
- **Features:**
- Self-learning systems
- Voice and image recognition
- Real-time processing
- **Examples:** AI assistants (Alexa, Siri), smart robots, quantum computers
- **Advantages:**
    - Intelligent decision making
    - Automation of complex tasks
    - IoT (Internet of Things) integration

Types Of Computer

a) Analog Computer:

- Works on continuous data (not digital).
- Used in scientific and engineering applications.
- Example: Speedometer, thermometer.

b) Digital Computer:

- Works on discrete data (0 and 1 – binary).
- Most common type of computer used today.
- Example: Laptop, mobile phone.

c) Hybrid Computer:

- Combination of analog and digital computer.
- Used in hospitals (e.g., ECG machines), scientific applications

Types Of Digital Computer

a) Microcomputer (Personal Computer – PC):

- Smallest and most commonly used.
- Designed for individual use.
- Example: Desktop, Laptop.

b) Mini Computer:

- Mid-sized computer.
- Supports multiple users at the same time.
- Used in small and medium organizations.

c) Mainframe Computer:

- Large and powerful systems.
- Can support hundreds or thousands of users simultaneously.
- Used in banks, railways, government departments.

d) Supercomputer:

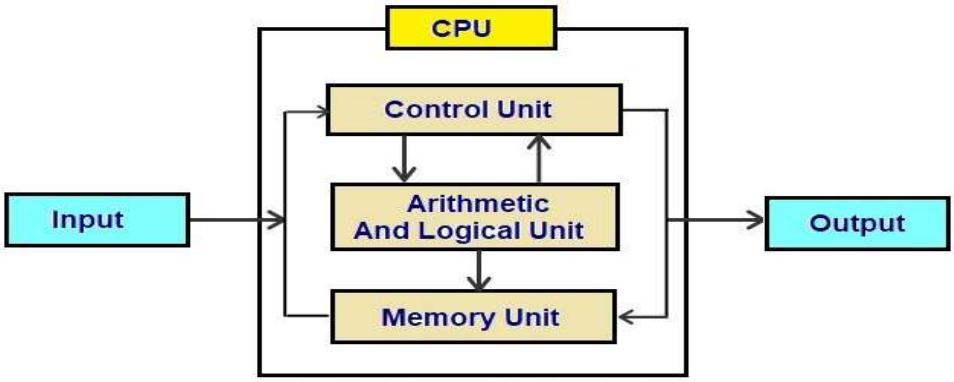
- Most powerful and fastest computer.
- Used in scientific research, weather forecasting, space research.
- Example: PARAM, Fugaku.

1.3 Basic Functions of a Computer

A computer generally performs five basic functions:

1. Input
2. Processing
3. Output
4. Storage
5. Control

Basic Computer Working



The control unit coordinates these activities.

1.3.1 Input

- **Input Devices** are hardware components that are used to **enter data and instructions** into the computer system for processing.
- These devices allow the user to interact with the computer by providing raw data, which is later processed by the CPU.

Functions of Input Devices:

- Accept data from the user.
- Convert human-readable data into machine-readable form (binary).
- Send data and instructions to the CPU for processing.

Examples of input devices:

- Keyboard, Mouse
- Scanner, Microphone
- Webcam, Touchscreen
- Barcode Reader

1.3.2 Processing

Processing means performing operations on input data to convert it into useful information.

Processing is mainly performed by the **CPU – Central Processing Unit**.

1.3.3 Output

- **Output Devices** are hardware components used to **display, print, or present the results** of data processing performed by the computer.
- They convert **machine-readable data** (binary) into **human-readable form** such as text, images, sound, or printed output.

Functions of Output Devices:

- Receive processed data from the CPU.
- Convert it into meaningful output.
- Present the result to the user in different forms (visual, audio, printed).
- Types of Output Devices:

Examples:

- Monitor, Printer
- Speaker, Projector

1.3.4 Storage

A computer stores data, programs and information for future use.

Examples:

- Hard Disk, SSD, Pen Drive, Memory Card
- CD/DVD, Cloud Storage

Storage can broadly be classified as:

Primary Storage

Examples:

- RAM
- ROM
- Cache Memory

Secondary Storage

Examples:

- HDD
- SSD
- Pen Drive
- Memory Card

1.3.5 Control

The computer must coordinate all operations.

The **Control Unit (CU)** of CPU controls and coordinates different activities of the computer.

Main Components of CPU

The CPU is mainly divided into **three parts**:

ALU (Arithmetic and Logic Unit)

- Performs all **arithmetic operations** like addition, subtraction, multiplication, division.
- Performs **logical operations** like comparison (greater than, less than, equal to).
- It is the **calculating unit** of the CPU.

CU (Control Unit)

- Manages and controls all activities inside the computer.
- It directs the flow of data between memory, ALU, and input/output devices.
- Does **not process** data itself but **controls** how other parts work.

Registers

- Small high-speed memory inside the CPU.
- Temporarily stores data, instructions, or results during processing.
- Faster than RAM and helps in **quick access** to frequently used data.

How CPU Works (In Simple Steps):

- **Fetch** – The control unit fetches instructions from memory.
- **Decode** – The instruction is decoded to understand the operation.

- **Execute** – ALU executes the instruction.
- **Store** – Result is stored in memory or sent to output.

Important Points:

- CPU is installed on the **motherboard**.
- The performance of a computer depends largely on the **speed and power of the CPU**.
- CPU speed is measured in **GHz (Gigahertz)**.
- Modern CPUs are **multi-core**, meaning multiple processing units in one chip (like Dual-core, Quad-core).

1.4 Characteristics of Computers

Computers have several important characteristics.

1. Speed

Computers can perform millions or billions of operations in a very short time.

For example, calculations that may take humans several minutes can be performed by a computer in fractions of a second.

2. Accuracy

Computers generally provide highly accurate results when:

- Data is correct
- Instructions/program are correct

This is often described as:

Garbage In, Garbage Out (GIGO)

If incorrect data is given to the computer, the output may also be incorrect.

Example:

```
Salary = ₹20,000
```

If the user mistakenly enters:

```
Salary = ₹2,00,000
```

the computer will calculate based on the incorrect input.

3. Storage Capacity

Computers can store large amounts of data.

For example, an organization can store:

- Employee records

- Customer information
- Sales records
- Financial records
- Inventory details

4. Diligence

A computer does not become tired or bored like humans.

It can perform the same operation repeatedly with consistent performance.

5. Automation

Once a program and required data are provided, a computer can perform tasks automatically.

Example:

A bank's system can automatically calculate:

- Interest
- Account balance
- Transaction records

6. Versatility

Computers can be used for many different purposes.

For example:

- Education, Banking
- Business, Healthcare
- Entertainment, Communication
- Manufacturing, Government services

7. Reliability

Computers can perform repetitive operations consistently.

However, reliability depends on:

- Hardware
- Software
- Correct input
- Proper maintenance

8. Multitasking

Modern computers can perform multiple tasks apparently at the same time.

For example, a user may:

- Play music

- Browse the Internet
- Download a file
- Edit a document

at the same time.

1.5 Limitations of Computers

Although computers are powerful, they also have limitations.

1. No Intelligence by Default

A computer does not independently understand a problem like a human unless appropriate software or AI systems are designed for it.

2. No Common Sense

Computers follow instructions and rules.

3. Dependency on Instructions

A computer needs instructions/programs to perform tasks.

4. Incorrect Input Produces Incorrect Results

This is known as:

GIGO – Garbage In, Garbage Out

5. No Emotions

Computers do not have human emotions or feelings.

6. Maintenance Requirement

Hardware and software require maintenance and updates.

1.6 Applications of Computers

Computers are used in almost every field.

Education

Computers are used for:

- Online classes, Digital learning
- Examination systems
- Student records, Presentations, Research

Banking

Computers are used for:

- ATM, Online banking
- Account management
- Fund transfer, Transaction processing

Business

Businesses use computers for:

- Accounting, Payroll
- Inventory, Sales
- Customer management
- Business analysis

Healthcare

Computers are used for:

- Patient records, Medical imaging
- Hospital management, Laboratory systems
- Appointment management

Government

Computers are used for:

- Digital records, Tax management
- Online services, E-governance, Public databases

Communication

Computers support:

- Email, Video conferencing
- Messaging, Social media, Internet communication

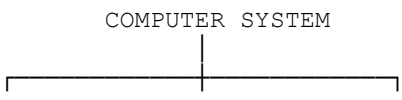
2. COMPONENTS OF COMPUTER SYSTEM

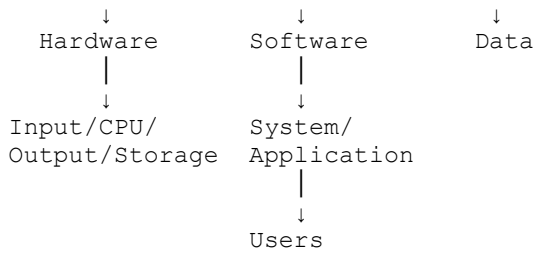
A computer system is not just a CPU or monitor. It consists of different components that work together.

The major components are:

1. Hardware
2. Software
3. Data
4. Users
5. Procedures

A simple representation is:





2.1 Hardware

Hardware refers to the physical parts of a computer system that can be seen and touched.

Examples:

- Keyboard, Mouse, Monitor, CPU
- Motherboard, RAM, Hard Disk, SSD
- Printer, Scanner, Speakers

Major Hardware Categories

Hardware can be broadly divided into:

1. Input devices
2. Processing devices
3. Output devices
4. Storage devices
5. Communication devices

2.2 Input Devices

Input devices are used to enter data and instructions into the computer.

Keyboard:

A keyboard is an input device used to enter:

- **Characters, Numbers, Symbols, Commands**

Mouse:

A mouse is a pointing device.

Common mouse operations:

- **Click, Double-click, Right-click, Drag and drop, Scroll**

Scanner:

A scanner converts physical documents or images into digital form.

Example:

A printed document can be scanned and stored as a digital file.

Microphone:

A microphone is used to input sound/audio into a computer.

Webcam:

A webcam captures images and video.

Applications include:

- Video conferencing
- Online classes
- Video calls

Barcode Reader:

A barcode reader reads information stored in barcodes.

Commonly used in:

- Retail stores
- Supermarkets
- Inventory systems

QR Code Scanner:

QR codes can store information such as:

- Website URLs
- Payment information
- Contact information
- Product information

2.3 Central Processing Unit – CPU

The **CPU (Central Processing Unit)** is responsible for executing instructions and performing processing operations.

It is often called the **brain of the computer**, although technically it is the main processing unit rather than a human-like brain.

The CPU mainly consists of:

1. **ALU – Arithmetic Logic Unit**
2. **CU – Control Unit**
3. **Registers**

Modern CPUs also contain additional components such as cache memory and multiple processing cores.

2.4 ALU – Arithmetic Logic Unit

ALU performs:

Arithmetic Operations

- Addition, Subtraction
- Multiplication, Division

Logical Operations

- Greater than, Less than, Equal to
- AND, OR, NOT

Example:

```
10 + 20 = 30
```

This calculation is handled by the processing unit, with arithmetic operations performed by the ALU.

2.5 Control Unit – CU

The **Control Unit** controls and coordinates the operations of the computer.

Its major responsibilities include:

- Fetching instructions
- Decoding instructions
- Controlling execution
- Coordinating CPU components
- Managing movement of data and instructions

The basic instruction cycle is:

```
Fetch → Decode → Execute
```

2.6 Registers

Registers are very small and very fast storage locations inside the CPU.

They temporarily hold:

- Data
- Instructions
- Addresses
- Intermediate results

Examples include:

- Accumulator
- Program Counter
- Instruction Register

- Memory Address Register

2.7 Memory

Memory is used to store:

- Data
- Instructions
- Programs
- Intermediate results

Memory is generally divided into:

Primary Memory

- RAM
- ROM
- Cache

Secondary Memory

- HDD, SSD
- Pen Drive, Memory Card
- Optical disks

2.8 Output Devices

Output devices present processed information to the user.

Examples:

- Monitor, Printer
- Speaker, Projector

Monitor

A monitor displays:

- Text, Images
- Videos, Graphics

Printer

A printer produces a physical copy of digital information.

This physical copy is called a **hard copy**.

Types include:

- Inkjet printer
- Laser printer
- Dot matrix printer

Speaker

A speaker produces audio output.

Projector

A projector displays computer output on a large screen.

It is commonly used in:

- Classrooms, Meetings
- Seminars, Conferences

2.9 Software

Software is a collection of programs and instructions that tells the computer hardware what to do.

Unlike hardware, software cannot normally be physically touched.

Examples:

- Windows, Linux
- MS Word, MS Excel
- Google Chrome
- VLC Media Player

Software can broadly be divided into:

1. System Software
2. Application Software

2.10 System Software

System software manages computer hardware and provides a platform for application software.

Examples:

- Operating Systems
- Device Drivers
- Utility Programs
- Language Translators

Examples of operating systems:

- Windows, Linux
- macOS, Android

2.11 Application Software

Application software is designed to perform specific tasks for users.

Examples:

Software	Purpose
MS Word	Document processing
MS Excel	Spreadsheet and calculations
PowerPoint	Presentation
Chrome	Web browsing
Photoshop	Image editing
Tally	Accounting

Application software can further be:

General Purpose Software

Used by many types of users.

Examples:

- Word processors
- Spreadsheets
- Presentation software

Special Purpose Software

Designed for a particular purpose.

Examples:

- Railway reservation system
- Banking software
- Hospital management system
- Payroll system

2.12 Users / Peopleware

People who use or operate computer systems are sometimes collectively referred to as **peopleware**.

Examples:

- Computer operators
- Programmers
- System analysts
- Database administrators
- Managers
- End users

2.13 Procedures

Procedures are the rules or steps that explain how a computer system should be used.

For example, in an organization:

```
Login
↓
Enter employee details
↓
Verify information
↓
Save record
↓
Generate report
```

2.14 Complete Computer System

Therefore, a computer-based system can be understood as:

```
Hardware
   +
Software
   +
Data
   +
People
   +
Procedures
   =
Computer-Based Information System
```

Advantages of Computer System

1. **High Speed** – Performs calculations and tasks very quickly.
2. **Accuracy** – Produces accurate results when given correct instructions.
3. **Storage** – Can store a large amount of data.
4. **Automation** – Performs tasks automatically.
5. **Multitasking** – Can perform multiple tasks at the same time.
6. **Reliability** – Can work continuously without getting tired.
7. **Easy Communication** – Helps in fast communication through email, internet, etc.

Disadvantages of Computer System

1. **Security Risks** – Data can be stolen or hacked.
2. **Health Problems** – Excessive use can cause eye strain and other health issues.
3. **Dependency** – Excessive dependence on computers can reduce human effort and skills.
4. **Job Displacement** – Automation can replace some human jobs.
5. **Cost** – Hardware, software, maintenance, and upgrades can be expensive.
6. **Data Loss** – Data may be lost due to hardware failure, viruses, or accidental deletion.
7. **Cybercrime** – Computers can be affected by viruses, malware, and other cyber attacks.

3. INDIAN COMPUTING ENVIRONMENT

The **Indian Computing Environment** refers to the development, adoption and use of computer technology, information technology and digital systems in India.

India has developed a large computing ecosystem involving:

- Government, Businesses, Educational institutions, Banks
- IT companies, Startups, Research organizations, Citizens

3.1 Development of Computing in India

The development of computing in India can be broadly understood in stages:

Early Stage (1940s to 1970s)

Computers were initially used mainly by:

- Government organizations
- Scientific institutions
- Research organizations
- Large businesses

Expansion Stage (1970s to 1990s)

With decreasing hardware costs and increasing availability of technology, computers became common in:

- Offices, Schools
- Colleges, Banks, Businesses

Internet Era (1990 to 2010)

The growth of the Internet introduced:

- Email, Websites, E-commerce
- Online banking, Digital communication

Digital Era (2010 to Present)

Modern India increasingly uses:

- Smartphones
- Cloud computing
- Digital payments
- Online government services
- E-commerce
- Artificial intelligence
- Data analytics
- Digital education

3.2 Role of Computers in India

Computers play an important role in:

Government

- E-governance, Digital records

- Online applications, Tax systems
- Public services

Banking

- ATM, Internet banking
- Mobile banking, Online transactions
- Digital payments

Education

- Online classes, Digital libraries
- Learning management systems
- Online examinations

Business

- Accounting, Marketing
- Customer management
- Inventory, E-commerce

Healthcare

- Digital patient records
- Hospital management
- Medical research
- Telemedicine

3.3 Digitalization in India

Digitalization means using digital technologies to improve services and processes.

Examples include:

- Online banking
- UPI-based payments
- Online ticket booking
- Digital government services
- Online education, E-commerce

Period	Stage	Major Development
1950–1970	Early Computing	Mainframes, TIFRAC, IBM computers
1970–1985	Computing Expansion	Business & government computerization
1985–1995	PC & IT Era	PCs, software, computer education, ERNET
1995–2005	Internet Expansion	ISP, cyber café, email, websites, VSNL public Internet
2005–2010	Broadband Era	Broadband, online services,Cable, Fibre Optics
2010–2016	Mobile Internet	3G, smartphones, apps
2015 onwards	Digital India	E-governance, digital services, e-Hospital

Period	Stage	Major Development
2016 onwards	Mass Internet	Jio, cheap mobile data, smartphones
2020 onwards	Digital Transformation	UPI, cloud, online education, remote work, digital businesses
2022 onwards	AI & Emerging Tech	AI, GenAI, cloud, IoT, 5G

3.4 Advantages of Computing Environment in India

1. Faster communication
2. Digital financial transactions
3. Better access to information
4. Improved government services
5. Growth of IT industry
6. Employment opportunities
7. Online education
8. Business automation
9. Better data management
10. Global connectivity

3.5 Challenges

India's computing environment also faces challenges such as:

- Cybersecurity threats
- Data privacy concerns
- Digital literacy gaps
- Unequal Internet access
- Infrastructure limitations
- Cyber fraud
- Need for skilled professionals

4. COMPUTERS IN BUSINESS

Computers have become an essential part of modern business organizations.

Businesses use computers to:

- Store information
- Process transactions
- Communicate
- Manage employees
- Maintain accounts
- Manage inventory
- Analyze sales
- Make decisions
- Communicate with customers

4.1 Major Applications of Computers in Business

1. Accounting

Computers are used for:

- Recording transactions
- Preparing invoices
- Calculating taxes
- Preparing financial statements
- Managing expenses

Example:

Accounting software can automatically calculate totals and generate reports.

2. Payroll Management

Businesses use computer systems to calculate:

- Employee salary
- Attendance
- Overtime
- Bonuses
- Deductions
- Taxes

3. Inventory Management

Computerized inventory systems track:

- Stock available, Purchases
- Sales, Stock movement
- Reorder levels

Example:

If inventory falls below a predefined level, the system can generate an alert.

4. Sales Management

Computers help businesses maintain:

- Sales records, Customer information
- Product information
- Invoices, Sales reports

5. Marketing

Computers are used for:

- Digital marketing
- Customer analysis
- Email marketing

- Social media marketing
- Advertising
- Market research

6. Human Resource Management

HR departments use computer systems for:

- Employee records, Recruitment
- Attendance, Payroll, Leave management
- Performance records

7. Customer Relationship Management

CRM systems help businesses manage customer information and interactions.

They can store:

- Customer names, Contact details
- Purchases, Complaints
- Communication history

8. Decision Making

Managers can use computer-generated reports to make business decisions.

For example:

```
Sales Report
     ↓
Data Analysis
     ↓
Identify Best-Selling Product
     ↓
Business Decision
```

4.2 Advantages of Computers in Business

Speed

Large amounts of business data can be processed quickly.

Accuracy

Computerized calculations reduce manual calculation errors.

Cost Reduction

Automation can reduce repetitive manual work.

Better Record Keeping

Data can be stored and retrieved easily.

Better Communication

Email, messaging and video conferencing improve communication.

Better Decision Making

Reports and data analysis provide useful information to managers.

Automation

Routine tasks can be automated.

5. MANAGEMENT OF DATA PROCESSING SYSTEMS IN BUSINESS ORGANIZATIONS

This topic is important because it connects **Computer Applications with Business Administration**.

5.1 What is Data Processing?

Data processing is the process of collecting raw data and converting it into meaningful information.

Basic Data Processing Cycle:

```
Data
 ↓
Input
 ↓
Processing
 ↓
Output
 ↓
Information
 ↓
Storage
```

Example:

A business receives the following sales data:

```
Product A → ₹10,000
Product B → ₹20,000
Product C → ₹15,000
```

After processing:

```
Total Sales = ₹45,000
```

The total sales figure is useful information for management.

5.2 Data Processing System

A **Data Processing System** is a combination of people, hardware, software, procedures and data used to collect, process, store and produce information.

5.3 Data Processing in Business

A business generates huge amounts of data every day.

Examples:

- Sales transactions
- Purchase records
- Employee information
- Customer records
- Inventory data
- Financial transactions

This data must be properly managed.

5.4 Steps in Data Processing

1. Data Collection

Data is collected from different sources.

Example:

Customer purchases.

2. Data Input

Collected data is entered into the computer system.

3. Data Processing

The system performs calculations, sorting, classification etc.

4. Output

Useful information is generated.

5. Storage

Information is stored for future use.

6. Distribution

Reports may be distributed to managers or departments.

5.5 Methods of Data Processing

The major methods of data processing are:

1. Manual Data Processing
2. Mechanical Data Processing
3. Electronic Data Processing

5.5.1 Manual Data Processing

Manual Data Processing is a method in which data is processed **by human beings without using computers or electronic machines**.

Example

A teacher manually adds the marks of students and calculates their average.

Advantages

- Simple to perform
- Low initial cost
- Suitable for small amounts of data

Disadvantages

- Time-consuming
- More chances of human error
- Difficult to handle large amounts of data
- Repetitive work becomes difficult

5.5.2 Mechanical Data Processing

Mechanical Data Processing uses **mechanical devices or machines** to perform data processing operations.

Examples

- Calculator
- Adding Machine
- Mechanical Accounting Machine
- Tabulating Machine

Example

A shopkeeper uses a calculator to calculate the total amount of daily sales.

Advantages

- Faster than manual processing
- Reduces some human effort
- Useful for simple calculations

Disadvantages

- Limited processing capabilities
- Less flexible

- Cannot handle complex tasks like modern computers

5.5.3 Electronic Data Processing (EDP)

Electronic Data Processing (EDP) is the process of using electronic computers and computer systems to collect, process, store, and retrieve data.

Electronic Data Processing is the processing of data using electronic computers and computer-based systems.

Example

In a college:

Student Marks → Computer → Processing → Result → Database

The computer can automatically calculate:

- Total marks, Percentage
- Grade, Pass/Fail status

Advantages of EDP

- Very fast, High accuracy
- Handles large amounts of data
- Reduces manual effort
- Data can be stored easily
- Reports can be generated quickly
- Repetitive tasks can be automated

Disadvantages of EDP

- **High Cost** – Requires computers, software, and infrastructure.
- **Skilled Staff Required** – Needs trained personnel.
- **Security Risks** – Data may be hacked or misused.
- **System Failure** – Hardware or software failure can stop work.
- **Power Dependency** – Requires continuous electricity.
- **Maintenance Required** – Systems need regular maintenance and updates.
- **Data Loss** – Data may be lost without proper backup.

5.6. Modes of Data Processing

The **mode of data processing** describes **how and when data is processed**.

The common modes are:

1. Batch Processing
2. Online Processing
3. Real-Time Processing

5.6.1 Batch Processing

Batch Processing is a method in which data is collected and stored over a period of time and then processed together as a group or batch.

Example: Salary Processing

A company has 1,000 employees.
Employee salary data is collected throughout the month.
At the end of the month, all salary data is processed together.

Data Collection → Batch Formation → Processing → Salary Reports

Examples

- Monthly salary processing
- Electricity bill generation
- Bank statements
- Examination results
- Monthly invoices

Advantages

- Suitable for large amounts of data
- Efficient for repetitive tasks
- Can process large groups of transactions

Disadvantages

- Results are not available immediately
- Errors may be discovered only after processing
- Not suitable for applications requiring immediate responses

5.6.2 Online Processing

Online Processing is a method in which data is entered into a computer system through an online/connected system and is processed without waiting for a large batch to be formed.

Examples

- Online shopping
- Online ticket booking
- Online banking
- ATM transactions

Example

When you purchase a product online:

Select Product → Enter Details → Make Payment → Order Confirmation

The system processes the transaction while you are using the service.

5.6.3 Real-Time Processing

Real-Time Processing is a method in which data is processed immediately after it is received, and the system provides an immediate or almost immediate response.

Example: ATM

You withdraw ₹5,000 from an ATM.

The system immediately:

1. Checks your account.
2. Checks your balance.
3. Approves the transaction.
4. Deducts ₹5,000.
5. Updates your account balance.

Examples

- ATM systems
- Online payment systems
- Railway reservation systems
- Airline reservation systems
- Stock market systems
- Industrial control systems

Batch vs Real-Time

Batch Processing	Real-Time Processing
Data is collected first	Data is processed immediately
Processing occurs later	Processing occurs as data arrives
No immediate response	Immediate response
Monthly salary processing	ATM transaction
Electricity bill generation	Online payment

5.7 Types of Data Processing

The classification of **types of data processing** can vary between textbooks. In computer applications, two important approaches for managing data are:

1. File Processing
2. Database Processing

Electronic Data Processing is also commonly discussed as a major form of computer-based data processing.

5.7.1 File Processing

File Processing is a method in which data is stored and managed in separate files.

Each application may maintain its own files.

Example

A college may have:

Student.txt

Contains:

- Roll Number, Name, Course

Marks.txt

Contains:

- Roll Number, Subject, Marks

Attendance.txt

Contains:

- Roll Number, Attendance

These are separate files.

Advantages

- Simple to understand
- Easy to implement for small applications
- Suitable for simple data requirements

Disadvantages

- Data duplication
- Data inconsistency
- Difficult data sharing
- Security problems
- Difficult maintenance
- Difficult to manage relationships between data

5.7.2 Database Processing

Database Processing is a method in which data is stored, organized, and managed in a centralized database.

A **Database Management System (DBMS)** is used to manage the data.

Examples of DBMS

- MySQL
- Oracle Database
- PostgreSQL
- Microsoft SQL Server
- MongoDB

Example

A college can maintain a central database containing:

Student Table

Roll No.	Name	Course
101	Rahul	BBA
102	Aman	BBA
103	Priya	BBA

Marks Table

Roll No.	Subject	Marks
101	Computer	85
102	Computer	78
103	Computer	92

Different applications can access the required information from the database.

Advantages

- Reduces data redundancy
- Improves data consistency
- Easy data sharing
- Better security
- Easy searching and updating
- Supports relationships between data
- Suitable for large applications

5.8. File Processing vs Database Processing

File Processing	Database Processing
Data is stored in separate files	Data is stored in a database

File Processing	Database Processing
Data duplication may be high	Data duplication can be reduced
Data sharing is difficult	Data sharing is easier
Security may be limited	Better security mechanisms
Relationships are difficult to manage	Relationships can be easily managed
Suitable for small applications	Suitable for large applications
DBMS is generally not required	DBMS is used

5.9 Important Difference: Method, Mode and Type

Students often get confused between these three terms.

Method of Data Processing

It describes **the basic technique used to process data**.

Examples:

- Manual
- Mechanical
- Electronic

Mode of Data Processing

It describes **when/how the data is processed**.

Examples:

- Batch
- Online
- Real-Time

Type/Approach of Data Processing

It can describe **how data is organized and managed**.

Examples:

- File Processing
- Database Processing

5.10 Role of Management in Data Processing

Management needs to ensure:

- Correct data collection
- Data security
- Proper storage
- Regular backups
- Access control
- Data accuracy
- Timely reports
- Efficient use of resources

6. PROGRAM DEVELOPMENT CYCLE

6.1 What is a Program?

A **program** is a set of instructions written in a programming language that tells a computer how to perform a particular task.

Example

Suppose we want to calculate the area of a rectangle.

The program needs to:

```
Input length
Input breadth
Calculate area = length × breadth
Display area
```

A program is a sequence of instructions written in a programming language to solve a particular problem or perform a specific task.

6.2 What is Program Development?

Program development is the systematic process of converting a problem into a working computer program.

It is not simply writing code.

A programmer generally has to:

1. Understand the problem
2. Analyze the requirements
3. Design the solution
4. Develop an algorithm
5. Prepare a flowchart or pseudocode
6. Write the program
7. Compile and test it
8. Remove errors
9. Document it
10. Maintain and update it

6.3 Program Development Cycle

The **Program Development Cycle** is a sequence of steps followed to develop a computer program systematically.

Main Steps

```
Problem Definition
        ↓
Problem Analysis
        ↓
Solution Design
        ↓
Algorithm / Pseudocode
        ↓
Flowchart
        ↓
Coding
        ↓
Compilation
        ↓
Testing & Debugging
        ↓
Documentation
        ↓
Implementation
        ↓
Maintenance
```

Let's understand every stage.

6.4 Step 1 – Problem Definition

The first step is to clearly understand and define the problem.

The programmer should determine:

- What problem needs to be solved?
- What is the objective?
- What information is required?
- What should the final output be?

Example

Problem: Write a program to calculate the average marks of three subjects.

Problem definition:

- Input: Marks of three subjects
- Processing: Add marks and divide by 3
- Output: Average marks

A clearly defined problem makes the remaining development process easier.

6.5 Step 2 – Problem Analysis

After defining the problem, the programmer analyzes it in detail.

The programmer identifies:

- Required inputs
- Required outputs
- Processing requirements
- Conditions
- Calculations
- Data requirements
- Possible constraints

Example

Problem: Find the largest of two numbers.

Analysis:

Input:

```
Number 1
Number 2
```

Processing:

Compare the two numbers.

Output:

Largest number.

6.6 Step 3 – Solution Design

In this stage, the programmer designs the logical solution.

Common tools used are:

- Algorithm
- Flowchart
- Pseudocode
- IPO Chart

These tools help the programmer understand the solution before writing actual code.

6.7 Step 4 – Algorithm Development

An **algorithm** is a finite sequence of well-defined steps used to solve a problem.

Example: Algorithm to Add Two Numbers

```
Step 1: Start
Step 2: Read A and B
Step 3: Calculate Sum = A + B
Step 4: Display Sum
```

```
Step 5: Stop
```

Characteristics of a Good Algorithm

A good algorithm should have:

1. **Input** – It may accept required input.
2. **Output** – It should produce a result.
3. **Definiteness** – Every step should be clear.
4. **Finiteness** – It should terminate after a finite number of steps.
5. **Effectiveness** – Each step should be practical and executable.

6.8 Step 5 – Flowchart

A **flowchart** is a graphical representation of an algorithm or process.

It uses standard symbols to represent different operations.

Example:

```
   START
     ↓
  Input A, B
     ↓
  Sum = A + B
     ↓
 Display Sum
     ↓
    STOP
```

Flowcharts will be discussed in detail in the next section.

6.9 Step 6 – Coding

After designing the solution, the programmer converts the algorithm into a programming language.

For example, in C++:

```
int a, b;
cin >> a >> b;

int sum = a + b;

cout << sum;
```

This process is called **coding** or **program implementation**.

6.10 Step 7 – Compilation

A compiler translates a high-level language program into machine-understandable form.

For example:

```
C++ Source Code
       ↓
```

```
    Compiler
       ↓
Object/Machine Code
```

6.11 Step 8 – Testing

Testing means checking whether the program works correctly for different inputs and situations.

For example, if we have a program to calculate division:

```
10 / 2 = 5
```

We should also test:

```
10 / 0
```

because division by zero is an invalid operation.

6.12 Step 9 – Debugging

Debugging is the process of finding and correcting errors in a program.

The errors are commonly called **bugs**.

Example

Suppose the programmer writes:

```
sum = a - b;
```

instead of:

```
sum = a + b;
```

The program may run successfully but produce the wrong result.

The programmer must identify and correct the error.

That process is debugging.

6.13 Types of Programming Errors

This is very important for exams.

1. Syntax Error

A syntax error occurs when the rules of the programming language are violated.

Example:

```
cout << "Hello"
```

If the required semicolon is missing, the compiler may report an error.

2. Logical Error

A logical error occurs when the program's logic is incorrect.

Example:

```
average = total / 5;
```

when there are actually 4 subjects.

The program may execute, but the result is incorrect.

3. Runtime Error

A runtime error occurs while the program is executing.

Example:

- Division by zero
- Invalid memory access
- File not found
- Insufficient resources

6.14 Step 10 – Documentation

Documentation means preparing information that explains the program and its operation.

Documentation can include:

- Program purpose
- Input requirements
- Output details
- Algorithm
- Flowchart
- Source code explanation
- User instructions
- Technical information

Documentation is useful for future maintenance and understanding.

6.15 Step 11 – Implementation

After successful testing, the program is deployed or implemented for actual use.

Example:

A company's payroll program is installed and made available to the HR department.

6.16 Step 12 – Maintenance

After implementation, programs may need changes.

Maintenance may be required to:

- Fix bugs
- Improve performance
- Add features
- Adapt to new hardware/software
- Improve security
- Modify existing functionality

Therefore:

Program development does not necessarily end after implementation.

7. FLOW CHARTING

7.1 What is a Flowchart?

A **flowchart** is a graphical representation of the steps involved in solving a problem or performing a process.

Instead of writing all steps in text, we represent them using standard symbols connected by arrows.

A flowchart is a diagrammatic representation of an algorithm or process using standard graphical symbols.

7.2 Why are Flowcharts Used?

Flowcharts help programmers:

- Understand a problem
- Visualize program logic
- Identify errors
- Communicate the solution
- Explain a program to others
- Plan the program before coding
- Document the logic

7.3 Standard Flowchart Symbols

Students should remember these symbols carefully because **flowchart symbol questions are common in examinations.**

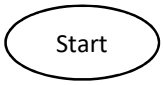
Symbol	Name	Purpose
Oval	Terminal	Start/Stop
Rectangle	Process	Calculation/processing
Parallelogram	Input/Output	Input or output
Diamond	Decision	Condition/decision
Arrow	Flow line	Direction of flow
Circle	Connector	Connects flowchart sections

7.4 Terminal Symbol

The terminal symbol represents:

- Start
- End
- Stop

Example:



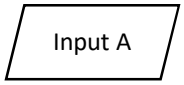
Usually represented by an **oval**.

7.5 Input / Output Symbol

The parallelogram is used for:

- Taking input
- Displaying output

Examples:



7.6 Process Symbol

A rectangle represents a processing operation.

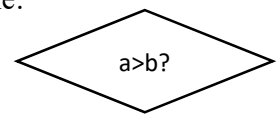
Example:

```
┌──────────────┐
│ Sum = A + B  │
└──────────────┘
```

7.7 Decision Symbol

A diamond represents a decision or condition.

Example:



A decision normally has two or more possible paths.

7.8 Flow Line

An arrow shows the direction in which the process moves.

Example:

```
START -> INPUT -> PROCESS -> OUTPUT -> STOP
```

7.9 Basic Flowchart Rules

When creating a flowchart:

1. Start with a **Start** symbol.
2. End with a **Stop/End** symbol.
3. Use standard symbols.
4. Connect symbols using arrows.
5. Maintain a logical flow.
6. Avoid unnecessary crossing of lines.
7. Decision paths should be clearly labelled.
8. Keep the flowchart simple and readable.
9. Use meaningful statements inside symbols.
10. Follow a consistent direction wherever possible.

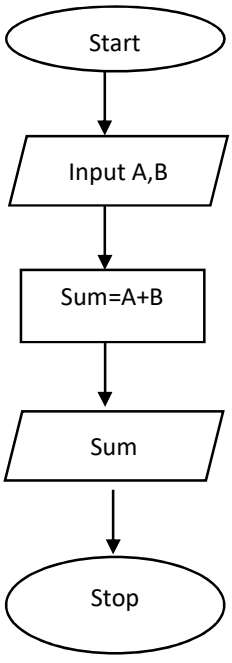
7.10 Example – Flowchart for Adding Two Numbers

Algorithm

```
Step 1: Start
Step 2: Input A and B
Step 3: Calculate Sum = A + B
Step 4: Display Sum
Step 5: Stop
```

Flowchart Logic

```
      START
        ↓
  Input A, B
        ↓
  Sum = A + B
        ↓
  Display Sum
        ↓
      STOP
```

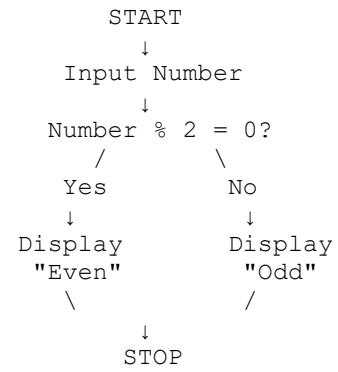


7.11 Flowchart for Finding Even or Odd Number

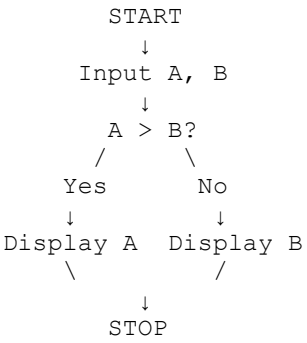
Logic

1. Start
2. Input number
3. Check whether number is divisible by 2
4. If yes → Even
5. Otherwise → Odd
6. Stop

Flowchart



7.12 Flowchart for Finding Largest of Two Numbers



7.13 Advantages of Flowcharts

1. Easy to understand
2. Helps in problem analysis
3. Improves communication
4. Helps identify logical errors
5. Useful before coding
6. Helps in documentation
7. Makes complex processes easier to visualize

7.14 Limitations of Flowcharts

1. Time-consuming for complex programs.
2. Large flowcharts can become difficult to manage.
3. Changes may require redrawing the chart.

4. Complex programs may require very large diagrams.
5. Maintaining flowcharts can be difficult.

8. INPUT-OUTPUT ANALYSIS

This topic is often represented as **IPO Analysis**.

IPO = Input – Process – Output

It is one of the simplest and most useful ways to analyze a programming problem.

8.1 What is Input?

Input is the data provided to a computer program.

Example:

For calculating the area of a rectangle:

```
Length
Breadth
```

are inputs.

8.2 What is Process?

Process refers to the operations performed on the input.

For rectangle:

```
Area = Length × Breadth
```

is the processing.

8.3 What is Output?

Output is the result produced by the program.

```
Area of Rectangle
```

is the output.

8.4 IPO Chart

An IPO chart represents a problem using three parts:

Input	Process	Output
Length, Breadth	Area = Length × Breadth	Area

8.5 Example – Addition of Two Numbers

Input	Process	Output
A, B	Sum = A + B	Sum

8.6 Example – Calculate Percentage

Suppose there are five subjects.

Input

```
Marks of 5 subjects
```

Process

```
Total = M1 + M2 + M3 + M4 + M5

Percentage = Total / 5
```

Output

```
Total Marks
Percentage
```

8.7 Importance of IPO Analysis

IPO analysis helps programmers:

- Understand the problem
- Identify required data
- Determine processing requirements
- Identify expected results
- Prepare algorithms
- Prepare flowcharts
- Write programs more efficiently

9. PROGRAMMING CONCEPTS

9.1 What is Programming?

Programming is the process of designing and writing instructions that a computer can execute to solve a problem.

A person who writes computer programs is called a **programmer**.

9.2 Programming Language

A **programming language** is a formal language used to write instructions for computers.

Examples:

- C
- C++
- Java

- Python
- JavaScript

Programming languages allow humans to express computational logic in a structured way.

9.3 Types of Programming Languages

At a basic level, programming languages can be classified as:

1. Machine Language
2. Assembly Language
3. High-Level Language

9.4 Machine Language

Machine language consists of binary instructions represented using:

```
0 and 1
```

It is directly understood by the computer's processor.

Advantages

- Directly executable by the computer
- Very fast execution

Disadvantages

- Difficult for humans to write
- Difficult to understand
- Difficult to debug
- Machine dependent

9.5 Assembly Language

Assembly language uses symbolic instructions called **mnemonics**.

Examples:

```
ADD
SUB
MOV
LOAD
```

An **assembler** translates assembly language into machine code.

9.6 High-Level Language

High-level languages are designed to be easier for humans to understand and write.

Examples:

- C

- C++
- Java
- Python

Example:

```
sum = a + b;
```

is much easier for a human to understand than binary machine instructions.

9.7 Compiler

A **compiler** translates a high-level language program into machine/object code, generally as a whole program or major translation unit depending on the language and toolchain.

Examples of commonly compiled languages include:

- C
- C++
- Go
- Rust

9.8 Interpreter

An **interpreter** executes a program by translating and executing instructions during runtime rather than producing a traditional standalone machine-code executable in the same manner as a compiler.

Python is commonly taught as an interpreted language, although modern Python implementations also perform compilation to intermediate bytecode.

9.9 Assembler

An **assembler** translates assembly language into machine/object code.

```
Assembly Language
        ↓
    Assembler
        ↓
Machine/Object Code
```

9.10 Algorithm vs Program

Algorithm	Program
Step-by-step solution	Actual coded implementation
Language independent	Written in a programming language
Easier to understand logically	Requires programming syntax
Developed before coding	Created during implementation

9.11 Algorithm vs Flowchart

Algorithm	Flowchart
Written representation	Graphical representation
Uses statements/steps	Uses symbols
Easy to write	Can take more time to draw
Easier to modify	Modification may require redrawing
Does not require graphical symbols	Uses standard symbols

9.12 Programming Concepts

Variable

A variable is a named memory location used to store a value that may change during program execution.

Example:

```
int age = 20;
```

Constant

A constant is a value that is intended not to change during program execution.

Example:

```
const float PI = 3.14159;
```

Data Type

A data type specifies the type of data a variable can store.

Examples:

- Integer
- Character
- Floating-point
- Boolean

Operator

An operator performs an operation on one or more operands.

Examples:

```
+, -, *, /, %, >, <, ==
```

Expression

An expression is a combination of values, variables and operators that produces a result.

Example:

```
a + b * 2
```

Statement

A statement is an instruction executed by a program.

Example:

```
cout << "Hello";
```

10. SOFTWARE DEVELOPMENT PROCESS

10.1 What is Software?

Software is a collection of programs, instructions and related components that enable a computer system to perform tasks.

Examples:

- Windows
- MS Word
- MS Excel
- Chrome
- Tally
- Photoshop

10.2 What is Software Development?

Software development is the process of:

Planning, designing, creating, testing, deploying and maintaining software.

Software development is broader than simply writing code.

10.3 Software Development Life Cycle – SDLC

SDLC stands for Software Development Life Cycle.

It provides a systematic approach to developing software.

A general SDLC includes:

```
Planning
   ↓
Requirement Analysis
   ↓
System Design
   ↓
Development / Coding
   ↓
Testing
   ↓
Deployment
   ↓
Maintenance
```

10.4 Phase 1 – Planning

The organization determines:

- What software is required?
- Why is it required?
- Who will use it?
- What resources are required?
- What is the expected cost?
- What is the expected timeline?

10.5 Phase 2 – Requirement Analysis

The development team collects and analyzes requirements from users and stakeholders.

Requirements may include:

- Functional requirements
- Performance requirements
- Security requirements
- User interface requirements
- Hardware/software requirements

Example

For a college management system:

Functional requirements may include:

- Student registration
- Attendance
- Fee management
- Examination records
- Reports

10.6 Phase 3 – System Design

In this phase, developers decide how the system will be built.

Design may include:

- System architecture
- Database design
- User interface
- Modules
- Data flow
- Security design

10.7 Phase 4 – Development

The actual software is developed by writing code.

Different programmers may work on different modules.

Example:

```
Login Module
Student Module
Payment Module
Report Module
```

10.8 Phase 5 – Testing

Software is tested to identify defects and verify that requirements are satisfied.

Testing may include:

- Unit testing
- Integration testing
- System testing
- User acceptance testing

10.9 Phase 6 – Deployment

The software is released and installed in its target environment.

Example:

A company's new payroll system is deployed on its servers and made available to employees.

10.10 Phase 7 – Maintenance

After deployment, software may need:

- Bug fixes
- Security updates
- Performance improvements
- New features
- Compatibility updates

10.11 Major SDLC Models

Students should have basic knowledge of common development models.

1. Waterfall Model

The Waterfall model follows a sequential approach.

```
Requirements
     ↓
Design
     ↓
Development
     ↓
Testing
     ↓
```

```
Deployment
     ↓
Maintenance
```

Each phase generally follows the previous phase.

Advantages

- Simple to understand
- Clearly defined phases
- Easy to manage

Disadvantages

- Changes can be difficult later
- Testing occurs relatively late
- Less suitable for frequently changing requirements

10.12 Prototype Model

In the Prototype model, an early working model of the system is created.

```
Requirements
      ↓
Prototype
      ↓
User Feedback
      ↓
Improvement
      ↓
Final System
```

It is useful when requirements are unclear.

10.13 Spiral Model

The Spiral model combines iterative development with systematic risk analysis.

It repeatedly goes through cycles involving:

- Planning
- Risk analysis
- Development
- Evaluation

It is useful for large and complex projects where risk management is important.

10.14 Agile Development

Agile is an iterative approach in which software is developed in small increments with frequent feedback.

Instead of waiting until the entire project is completed, teams deliver smaller parts of the system regularly.

Major characteristics

- Iterative development
- Frequent feedback
- Customer collaboration
- Adaptability
- Continuous improvement

10.15 Software Development vs Program Development

This difference is useful for examinations.

Program Development	Software Development
Focuses mainly on developing a program	Broader process of developing software systems
Usually solves a specific problem	Can involve complete applications/systems
Often smaller in scope	Can be large and complex
Coding is a major component	Includes planning, requirements, design, coding, testing, deployment and maintenance